PAPER LOGIC MAP

How a paper builds its case, section by section

A worked analysis of J. Zhang, Y. Dou, G. Liu and Z. Guo, “First-principles study of stacking fault energies in Mg-based binary alloys”, *Computational Materials Science* 79 (2013) 564–569.

What this report is

This report reads one research paper as a chain of questions. Each section of a paper answers a small set of questions in a fixed order, and each answer creates the need for the next. The report maps those questions for the introduction, methods, results, discussion and conclusions, shows which pieces of the paper answer each one, and notes the writing move behind every piece.

It also follows the thread that ties the sections together: every gap named in the introduction should become a promise, every promise a method, every method a result, and every result a line in the conclusion. Where the thread breaks, or where the paper leaves out something a reader needs, the report says so. Those points are collected in a checklist at the end that you can apply to your own manuscript. Two appendices describe how the analysis was made and compare it with a divide-and-conquer approach to paper assessment.

The five-part chain

Section	Question it answers	Key move in this paper
Introduction	Why is this study needed?	Funnel from Mg lightness to the I_1 stacking fault, then two kinds of gap: coverage and method.
Methods	How exactly was it done?	A two-fault periodic supercell that needs no vacuum layer, validated by a convergence test.
Results	What was found?	52 solutes split into SFE raisers and lowerers; every rare earth lowers the SFE.
Discussion	Why does it happen?	Two descriptors, ionization energy and atomic radius, where the second explains the failures of the first.
Conclusions	What does the paper claim?	A one-line rule: low ionization energy and large radius give a low SFE.

How to read the maps

Each section opens with a map. Dark boxes down the middle are the questions, in order; a green edge marks a step where the paper delivers, a red edge a step where something is missing. Coloured boxes either side are pieces of the paper that answer that question. Dashed boxes are not in the paper: either an implied gap, or something a reader would need but cannot find. The pages after each map explain every step and node.



Why this study is needed

Eight questions, broad to narrow. Steps 1–5 establish the territory, step 6 names the gap, and steps 7–8 say what the paper does and why it matters.



Grey lines link each node to its step. Green arrows: a gap answered, a promise delivered, or a puzzle explained. Dashed blue arrows: an idea planted in one place and used in another.

1 · Introduction, paragraph 1 of 6

Why does the field matter?

Magnesium alloys are the lightest structural metals, so they are attractive wherever weight matters. That is the reason anyone should care about the rest of the paper.

> ***Writing move.*** *Open wide enough that a materials scientist outside the subfield sees the stakes, but no wider. One or two sentences is enough.*
>
> *Signal phrase: "being the lightest structural materials"*

NODE	IN THE PAPER	WRITING MOVE
CONCEPT OR CHOICE **Lightest structural metal** aerospace · automotive · electronics	Low density means lighter vehicles and aircraft and better fuel efficiency.	Application hooks give the reader a reason to care in one line.
CONCEPT OR CHOICE **Wrought use is still limited** sheets · extrusions	Despite the promise, sheet and extruded Mg products are uncommon.	A tension between promise and reality creates the first pull toward a problem.

2 · Introduction, paragraph 1 of 6

What is the specific problem?

Wrought Mg products are rare because hcp Mg deforms poorly at room temperature. The cause is a shortage of easy slip systems, so non-basal slip and twinning have to be activated.

> ***Writing move.*** *Turn the broad context into a concrete technical obstacle. The word "However" usually marks this turn.*
>
> *Signal phrase: "is still very limited"*

NODE	IN THE PAPER	WRITING MOVE
NUMBER OR SETTING **Only 2 independent basal slip systems** hcp Mg	In hcp Mg the basal plane supplies just two independent slip systems.	A hard number makes the problem concrete instead of vague.
NUMBER OR SETTING **5 independent modes needed** compatible deformation	Compatible polycrystal deformation needs five independent modes, so basal slip alone falls well short.	Pair the deficit with the requirement so the size of the problem is obvious.
CONCEPT OR CHOICE **Activate non-basal slip and twinning** [1] Feng & Ma 2009 · [2] Wu et al. 2008	The route to formability is switching on deformation modes outside the basal plane.	State what would solve the problem. The next paragraph names the lever that controls it.

3 · Introduction, paragraph 2 of 6

What controls the problem?

Stacking fault energy governs how a metal deforms, and alloying elements can change it by disturbing the local charge density and strain field.

> ***Writing move.*** *Introduce the one lever your paper is about. Define it briefly and show it can be tuned, which sets up a study of many solutes.*
>
> *Signal phrase: "It is well known that"*

NODE	IN THE PAPER	WRITING MOVE
CONCEPT OR CHOICE **SFE governs plastic deformation** stacking fault energy	SFE is a key parameter for deformation behaviour and mechanical properties in metals and alloys.	Introduce the central quantity once, in plain terms.
CONCEPT OR CHOICE **Alloying tunes SFE** [3] Datta et al. 2006	Solute atoms disturb the local charge density and strain field of the matrix, which changes SFE.	This mechanism reappears in the discussion as ionization energy and atomic radius, so planting it early pays off.
CITED SOURCE **Precedent in fcc Al and Cu** [4] Qu et al. 2009 · [5] Mishra et al. 2007	In fcc metals, dislocation activity, twinning and grain fragmentation depend strongly on SFE.	Borrow credibility from a better-studied system before turning to your own.

4 · Introduction, paragraph 3 of 6

Why believe it matters in Mg?

Three studies tie SFE to ductility and deformation mode in Mg alloys specifically, with Y, Zn, Li and Al.

> ***Writing move.*** *Use citations as evidence for a claim, one claim per study. Each reference earns its place by moving the argument forward.*
>
> *Signal phrase: "have indicated a close association"*

NODE	IN THE PAPER	WRITING MOVE
CONCEPT OR CHOICE **The I_1 basal stacking fault** intrinsic fault	I_1, bounded by pyramidal partials, acts as a nucleation site for <c+a> dislocations. It is the fault this paper computes.	Narrow from SFE in general to the exact defect studied, and justify that choice with evidence.
CITED SOURCE **Sandlöbes et al. 2012: Mg–Y** [6] Acta Mater. 60	Y lowers the I_1 SFE, which helps nucleate pyramidal <c+a> dislocations and raises ductility.	The strongest citation comes first and directly links the lever to the problem.
CITED SOURCE **Datta et al. 2008: Mg–Zn–Y** [7] Acta Mater. 56	Zn lowers SFE and stabilises a 6-layer structure in which prismatic slip is active.	A second, independent study shows the link is not a one-off.
CITED SOURCE **Han et al. 2011: Li vs Al** [8] Scripta Mater. 64	Li raises SFE and favours dislocation processes; Al lowers it and favours sequential faulting.	Showing solutes can push SFE either way motivates surveying many elements.

5 · Introduction, paragraph 4 of 6

How can it be studied?

Measured SFEs scatter widely, while DFT values for pure Mg agree much better. This justifies choosing DFT as the method.

> ***Writing move.*** *Compare the available tools and show why yours is the sensible choice before you criticise how others used it.*
>
> *Signal phrase:* *"Unfortunately, experimental measurements"*

NODE	IN THE PAPER	WRITING MOVE
NUMBER OR SETTING **Experiments: 50–280 mJ/m²** [9–11]	SFE measurements need delicate techniques and scatter over a very wide range.	Quantify the weakness of the alternative method instead of just calling it unreliable.
NUMBER OR SETTING **DFT on pure Mg: 8–21 mJ/m²** [16–19]	First-principles values for pure Mg fall in a much narrower band.	The contrast with the experimental range justifies the method in one comparison.

6 · Introduction, paragraph 5 of 6

What is still missing?

Few solutes have been computed for the I_1 fault, and earlier setups used vacuum layers whose free surfaces may interfere, with convergence rarely checked.

> ***Writing move.*** *State the gap plainly and make it follow from the steps above. Strong gaps combine a coverage problem with a method problem.*
>
> *Signal phrase:* *"are still very limited at present"*

NODE	IN THE PAPER	WRITING MOVE
GAP **Few solutes studied for I_1** Al, Dy, Gd, Li, Nd, Pr, Tb, Y [7,8,21]	I_2 has been computed for 18 solutes [8,20], but I_1 only for eight.	A coverage gap. Listing what exists makes the missing part visible.
GAP **Supercell convergence rarely tested** method gap	The paper argues that testing SFE against supercell size is essential for reliable values.	A rigour gap. It is phrased as a requirement rather than an attack on other authors.
GAP **Vacuum layers add free surfaces** method gap	Earlier calculations used vacuum to handle the image effect, which introduces surfaces that may interact with the fault.	A method gap, stated fairly: the interaction may be negligible, but only if checked.
GAP **Only one solute concentration** implicit gap	Not stated outright, but the contribution of two concentrations only makes sense if earlier work used one.	An unstated gap weakens the argument. Writing it out would make step 7 land harder.

7 · Introduction, paragraph 6 of 6

What do we do?

A two-fault periodic supercell removes the need for vacuum, convergence is tested, and 52 solutes are computed at two concentrations.

> ***Writing move.*** *Answer the gap point by point. Every weakness raised in step 6 should have a matching item here, and every item here needs a matching method detail. The methods map below follows this step.*
>
> *Signal phrase: "In this paper"*

NODE	IN THE PAPER	WRITING MOVE
THIS WORK **52 binary Mg–X systems** Table 2	The survey covers 52 solutes, from alkali metals to transition metals and rare earths.	Answers the coverage gap directly.
THIS WORK **Convergence test of supercell size** 1×1, 2×2, 3×3 × X cells	SFE is checked against supercell size; at least 10 unit cells (20 layers) along c are needed.	Answers the rigour gap and doubles as validation in the results.
THIS WORK **Two-fault periodic supercell** no vacuum layer	Each periodic cell holds two fault interfaces, so no free surface is introduced; the cell is made long along c instead.	Answers the vacuum-layer gap with a different philosophy, not just more computing.
THIS WORK **Two concentrations: 11 and 25 at.%** on the fault plane	One solute in a 3×3×10 or a 2×2×10 cell gives 11 or 25 at.% on the faulted layer.	Answers the implicit concentration gap and enables the later discussion of concentration effects.

8 · Introduction, paragraph 6 of 6

Why does it matter?

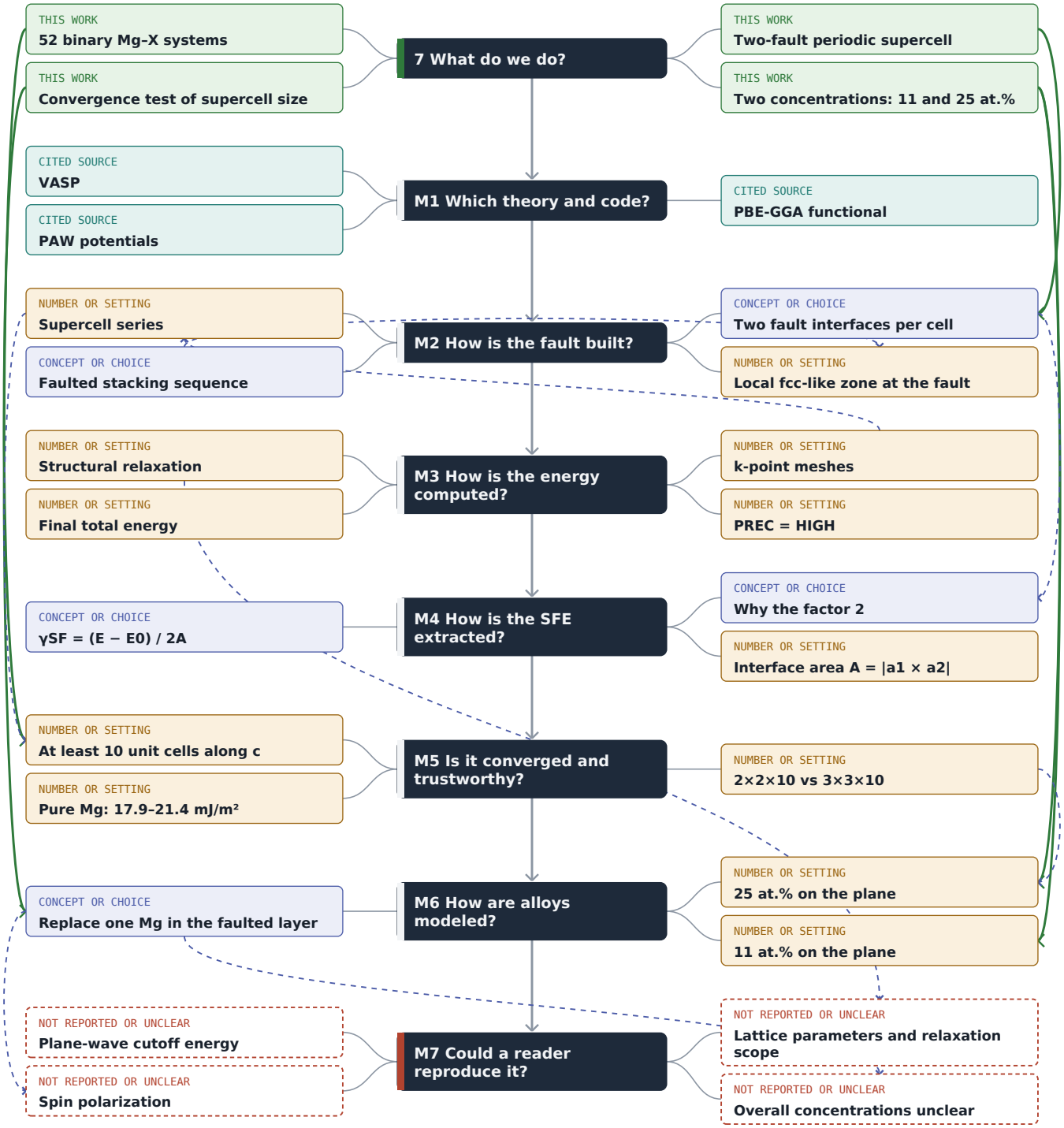
The results form a first-principles SFE database meant to guide alloy design, and the trends are explained with simple physical features.

> ***Writing move.*** *End on the use of the work. Many journals also expect a one-sentence preview of the main finding here, which this paper leaves to the abstract.*
>
> *Signal phrase: "expected to serve as a useful guideline"*

NODE	IN THE PAPER	WRITING MOVE
THIS WORK **A first-principles SFE database** guideline for alloy design	The values are meant as a reference for designing new Mg alloys.	The practical payoff ties back to the formability problem from step 2.
THIS WORK **Trends from physical features** ionization energy · atomic radius · tunneling	SFE trends are interpreted through simple properties of the solute atoms.	The scientific payoff previews the discussion section and closes the loop with the alloying mechanism from step 3.

How the promises are kept

The four promises from introduction step 7 sit at the top. Each green arrow points to the method detail that delivers it. The last step lists what a reader would still need to reproduce the calculations.



Grey lines link each node to its step. Green arrows: a gap answered, a promise delivered, or a puzzle explained. Dashed blue arrows: an idea planted in one place and used in another.

7 · Introduction, paragraph 6 of 6

What do we do?

The introduction promised four things. The methods section has to show how each one is carried out.

> ***Writing move.*** *Read your own "this work" paragraph as a checklist. Every promise needs a matching method detail; the green arrows show where each is delivered.*

NODE	IN THE PAPER	WRITING MOVE
THIS WORK **52 binary Mg–X systems** promise	The coverage promise: many solutes, not a handful.	Delivered by the substitution scheme in M6, applied to every element in Table 2.
THIS WORK **Convergence test of supercell size** promise	The rigour promise: show the result does not depend on cell size.	Delivered by the supercell series and the convergence result in M5.
THIS WORK **Two-fault periodic supercell** promise	The method promise: no vacuum layers and no free surfaces.	Delivered by the two-interface geometry in M2.
THIS WORK **Two concentrations: 11 and 25 at.%** promise	The concentration promise.	Delivered by the two in-plane cell sizes in M6, which M5 shows give comparable pure-Mg values.

M1 · Section 2, paragraph 1

Which theory and code?

Density functional theory in VASP, with projector augmented wave potentials and the PBE form of the generalized gradient approximation.

> ***Writing move.*** *Name the code, the theory level and the functional first, each with its standard citation. Readers judge the credibility of every later number from this line.*
>
> *Signal phrase: "calculations were implemented in"*

NODE	IN THE PAPER	WRITING MOVE
CITED SOURCE **VASP** [22–24] Kresse et al.	The Vienna Ab initio Simulation Package performs the calculations.	Cite the code papers, not just the name.
CITED SOURCE **PAW potentials** [25, 26] Kresse & Joubert · Blöchl	Core electrons are treated with the projector augmented wave method.	State the potential type; a reader also needs to know which valence configurations were used.
CITED SOURCE **PBE-GGA functional** [27] Perdew, Burke, Ernzerhof 1996	Exchange and correlation are approximated with PBE within the GGA.	The functional affects absolute SFE values, so it must always be named.

M2 · Section 2, paragraph 2, with Fig. 1

How is the fault built?

Supercells are stacked along c. The faulted cell switches from AB to CB stacking halfway, and periodic boundaries give two fault interfaces per cell.

> ***Writing move.*** *Describe the geometry well enough that someone could rebuild it. A figure of the normal and faulted cells does much of the work.*
>
> *Signal phrase: "A series of supercells"*

NODE	IN THE PAPER	WRITING MOVE
NUMBER OR SETTING **Supercell series** 1×1, 2×2, 3×3 in-plane × X = 4–12 along c	Fifteen cells are built by repeating the hcp unit cell, to test size effects in and out of plane.	Table 1 lists every cell with its atom count, so the reader sees the full test matrix.
CONCEPT OR CHOICE **Faulted stacking sequence** ABABABABAB|CBCBCBCBCB	Half the cell keeps AB stacking and half shifts to CB, creating the I_1 fault at the boundary.	Writing out the stacking sequence defines the defect unambiguously.
CONCEPT OR CHOICE **Two fault interfaces per cell** periodic boundary · no vacuum	Because the cell repeats along c, the AB|CB boundary also appears at the cell edge, giving two faults 10 layers apart in a 2×2×10 cell.	This is the paper's key methodological choice; it is stated clearly and shown in Fig. 1.
NUMBER OR SETTING **Local fcc-like zone at the fault** 5.178 Å vs 5.165 Å after relaxation	At each fault the stacking is locally fcc-like, and relaxation opens the interplanar distance slightly.	A small structural observation here sets up the atomic-radius argument in the discussion.

M3 · Section 2, paragraph 1

How is the energy computed?

Atoms relax with first-order Methfessel–Paxton smearing until forces fall below 0.01 eV/Å, then the total energy is taken with the tetrahedron method.

> ***Writing move.*** *List every setting that can change the number (smearing, force criterion, k-points, precision) together in one place.*

NODE	IN THE PAPER	WRITING MOVE
NUMBER OR SETTING **Structural relaxation** Methfessel–Paxton, σ = 0.2 eV · forces < 0.01 eV/Å [29]	Atomic positions are relaxed with first-order smearing until residual forces are small.	Give both the smearing method and the stopping criterion.
NUMBER OR SETTING **Final total energy** tetrahedron method + Blöchl correction [30]	After relaxation, energies are recomputed with the linear tetrahedron method for accuracy.	Separating the relaxation and energy steps shows care with metallic systems.
NUMBER OR SETTING **k-point meshes** Monkhorst–Pack [28] · 19×19×3 · 9×9×3 · 7×7×3	The in-plane mesh shrinks as the in-plane cell grows, keeping sampling density similar.	Report the mesh for every cell size, as Table 1 does.
NUMBER OR SETTING **PREC = HIGH** VASP precision setting	All calculations use high precision in the INCAR file.	Code-specific flags help other users of the same code reproduce the setup.

M4 · Section 2, paragraph 3, Eq. (1)

How is the SFE extracted?

SFE is the energy difference between the faulted and the perfect supercell, divided by twice the basal area.

> ***Writing move.*** *Give the defining equation and explain every symbol, including any factor that is not obvious, such as the 2 here.*
>
> *Signal phrase: "The SFE I_1 is extracted by"*

NODE	IN THE PAPER	WRITING MOVE
CONCEPT OR CHOICE **$\gamma_{SF} = (E - E_0) / 2A$** Eq. (1)	E is the faulted supercell energy and E_0 the perfect one; the difference is the cost of the faults.	Define each symbol right after the equation.
CONCEPT OR CHOICE **Why the factor 2** two interfaces per cell	The energy difference contains two faults, so it is split between them.	This links the equation back to the geometry choice in M2.
NUMBER OR SETTING **Interface area $A = \lvert a_1 \times a_2 \rvert$** basal cell area	The area is the size of the cross product of the in-plane lattice vectors.	Even simple definitions avoid ambiguity about which area is meant.

M5 · Section 3.1 with Table 1 and Fig. 3

Is it converged and trustworthy?

The SFE of pure Mg is computed in 15 supercells. Values settle once there are at least 10 unit cells along c and agree with published DFT values.

> ***Writing move.*** *Show the method works on a known case before applying it to new ones. This paper puts the test in Results 3.1, but logically it belongs to the method.*

NODE	IN THE PAPER	WRITING MOVE
NUMBER OR SETTING **At least 10 unit cells along c** ≥ 20 atomic layers	SFE stops changing once the two faults are 10 or more layers apart.	Turn the test into a rule a reader can reuse.
NUMBER OR SETTING **Pure Mg: 17.9–21.4 mJ/m²** agrees with [16–19]	Converged values fall inside the published DFT range of 8–21 mJ/m².	Benchmarking against the literature closes the loop opened in introduction step 5.
NUMBER OR SETTING **2×2×10 vs 3×3×10** 21.0 vs 21.4 mJ/m²	The two cells used for alloys give almost the same pure-Mg SFE.	This justifies comparing results from different cell sizes as concentration effects rather than size artefacts.

M6 · Section 2, paragraph 4, with Fig. 2

How are alloys modeled?

One Mg atom in the faulted layer is replaced by a solute. The in-plane size of the cell then sets the concentration on that plane.

> ***Writing move.*** *Explain how the variable of interest (here concentration) is controlled, and justify the choice using the validation step.*

NODE	IN THE PAPER	WRITING MOVE
CONCEPT OR CHOICE **Replace one Mg in the faulted layer** substitutional solute · Fig. 2	A single solute atom sits on the fault plane in each alloy cell.	Say exactly where the solute goes, since SFE depends strongly on its distance from the fault.
NUMBER OR SETTING **25 at.% on the plane** 2×2×10 · 1 of 4 atoms	A 2×2 basal layer holds four atoms, so one solute is a quarter of the layer.	Show the arithmetic behind each concentration.
NUMBER OR SETTING **11 at.% on the plane** 3×3×10 · 1 of 9 atoms	A 3×3 basal layer holds nine atoms, so one solute is about 11 % of the layer.	The lower concentration is the one used in the discussion, to limit solute–solute interaction.

M7 · Not in the paper: a reader's check

Could a reader reproduce it?

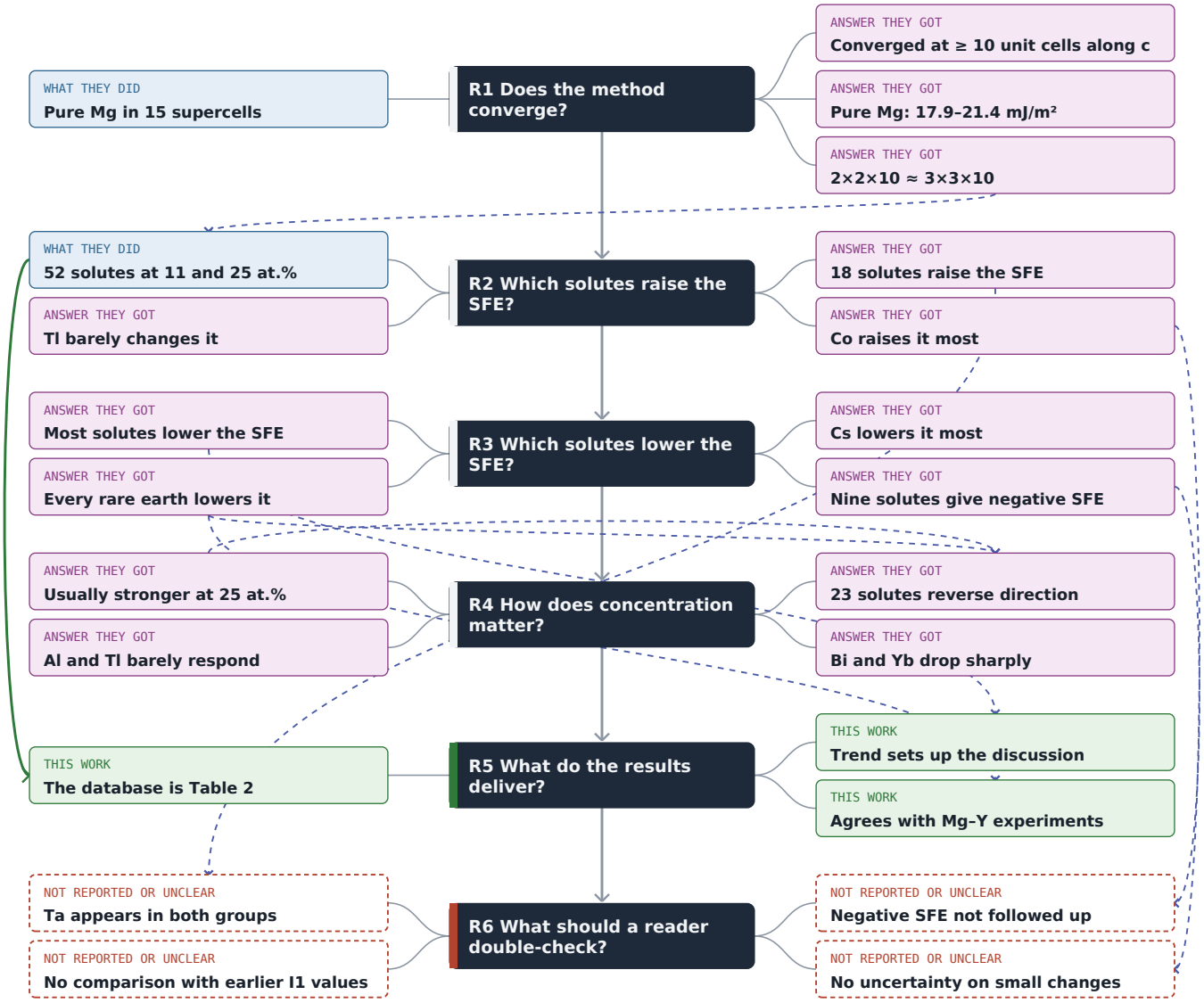
Several settings a reader would need to repeat these calculations are not reported.

> ***Writing move.*** *Before submitting, ask a colleague to rebuild your calculation from the methods alone. Anything they have to ask you for belongs in the text or the supplement.*

NODE	IN THE PAPER	WRITING MOVE
NOT REPORTED OR UNCLEAR **Plane-wave cutoff energy** ENCUT not given	The basis-set cutoff is not reported, although it directly affects total energies.	Always report the cutoff and how it was tested.
NOT REPORTED OR UNCLEAR **Spin polarization** Fe, Co, Ni, Mn, Cr can be magnetic	The text does not say whether calculations were spin-polarized. For magnetic solutes this can change the energy noticeably.	State the spin treatment whenever transition-metal solutes are involved.
NOT REPORTED OR UNCLEAR **Lattice parameters and relaxation scope** a, c/a · cell shape fixed or relaxed?	The Mg lattice constants used, and whether the cell shape or only atom positions were relaxed, are not given.	Report the equilibrium lattice constants and the degrees of freedom that were relaxed.
NOT REPORTED OR UNCLEAR **Overall concentrations unclear** 1/80 = 1.25 at.% · 1/180 = 0.56 at.%	The paper gives 2.5 and 1.1 at.% overall. Those match two solutes per cell, while the text describes replacing one atom. It may mean one per fault interface, but this is not stated.	Check that every number in the methods can be derived from the description.

What they found

Each step asks one question of the data. Blue nodes say what was computed and purple nodes give the answer. Concentration effects appear in the paper's discussion (§4.2) but are findings, so they are mapped here.



Grey lines link each node to its step. Green arrows: a gap answered, a promise delivered, or a puzzle explained. Dashed blue arrows: an idea planted in one place and used in another.

R1 · Section 3.1 with Table 1 and Fig. 3

Does the method converge?

Before any alloy, the authors check that pure Mg gives a stable, believable SFE in their supercells.

> ***Writing move.*** *Open the results with the validation, so every later number inherits its credibility. Keep it short: one table, one figure, one rule.*
>
> *Signal phrase: "It can be seen from Fig. 3"*

NODE	IN THE PAPER	WRITING MOVE
WHAT THEY DID **Pure Mg in 15 supercells** 1×1, 2×2, 3×3 in-plane × X = 4–12	The same pure-Mg fault is computed in every cell of the test series.	Say what was run before saying what came out.
ANSWER THEY GOT **Converged at ≥ 10 unit cells along c** Fig. 3 · ≥ 20 atomic layers	All three in-plane series level off once the faults are at least 10 layers apart.	State the answer as a reusable rule.
ANSWER THEY GOT **Pure Mg: 17.9–21.4 mJ/m²** within published DFT values [16–19]	The converged values agree with earlier calculations.	Agreement with literature is the evidence that the setup is right.
ANSWER THEY GOT **2×2×10 ≈ 3×3×10** 21.0 vs 21.4 mJ/m²	The two cells chosen for alloys give nearly the same pure-Mg SFE.	This one comparison licenses treating later differences as concentration effects.

R2 · Section 3.2 with Table 2

Which solutes raise the SFE?

Eighteen of the 52 solutes raise the SFE of Mg, cobalt by far the most.

> ***Writing move.*** *Report the main split in the data first (raise vs lower), name the extreme case with its numbers, and point to the table for the rest.*
>
> *Signal phrase: "As Table 2 shows"*

NODE	IN THE PAPER	WRITING MOVE
WHAT THEY DID **52 solutes at 11 and 25 at.%** one atom on the fault plane · 104 alloy cells	Each solute replaces one Mg atom in the faulted layer, in both cell sizes.	Remind the reader of the scope in one line before the findings.
ANSWER THEY GOT **18 solutes raise the SFE** Ag Be Co Cr Cu Fe Li Mn Mo Nb Ni Pd Re Rh Ru Ta V W	These are mostly transition metals, plus Be, Li and Ag.	A complete list in the text is fine when it is short; otherwise point to the table.
ANSWER THEY GOT **Co raises it most** 34.8 (11 at.%) · 54.1 mJ/m² (25 at.%)	Cobalt more than doubles the SFE at 25 at.% on the fault plane.	Naming the extreme case with numbers gives the reader the size of the effect.
ANSWER THEY GOT **Tl barely changes it** 19.7 · 19.5 mJ/m²	Thallium leaves the SFE close to the pure-Mg value at both concentrations.	A neutral case helps define the boundary between the two groups.

R3 · Section 3.2 with Table 2

Which solutes lower the SFE?

Most solutes lower the SFE. Cesium lowers it most, nine solutes push it below zero, and every rare earth lowers it.

> ***Writing move.*** *Group results into patterns a reader can remember (all rare earths, the negative-SFE group) rather than walking through the table row by row.*
>
> *Signal phrase: "majority of alloying elements decrease the SFE"*

NODE	IN THE PAPER	WRITING MOVE
ANSWER THEY GOT **Most solutes lower the SFE** incl. Al, Ca, Sr, Zn, Zr, Sn	The lowering group includes the elements most used in commercial Mg alloys.	Linking a result to practice makes it matter to a wider audience.
ANSWER THEY GOT **Every rare earth lowers it** Y, Gd, Nd, Dy, Er, Ce, La, Sm, Tb …	All rare-earth solutes studied lower the SFE, with no exception.	An exceptionless pattern is the strongest kind of result; the authors flag it as pointing to an underlying mechanism.
ANSWER THEY GOT **Cs lowers it most** down to −33.3 mJ/m² at 25 at.%	Cesium has the largest effect of all solutes, in the lowering direction.	Pair the two extremes (Co and Cs) so the reader sees the full range.
ANSWER THEY GOT **Nine solutes give negative SFE** Ba Ce Cs K La Nd Pr Rb Yb	A negative SFE would mean the faulted stacking is more stable than the perfect one in these cells.	The authors present this carefully as a hint, not a claim.

R4 · Section 4.2 with Fig. 7 (in the discussion)

How does concentration matter?

For most solutes the effect grows from 11 to 25 at.%, but 23 solutes reverse direction at the higher concentration.

> ***Writing move.*** *This is a result, yet the paper places it in the discussion. Keeping all findings in the results section and all explanations in the discussion makes a paper easier to follow.*

NODE	IN THE PAPER	WRITING MOVE
ANSWER THEY GOT **Usually stronger at 25 at.%** Ga: 19.0 → 3.3 mJ/m²	For most solutes, more solute on the fault plane means a bigger change in SFE.	Give the expected trend first, with one clear example.
ANSWER THEY GOT **Al and Tl barely respond** Fig. 7, black lines	Raising the concentration of Al or Tl changes the SFE very little.	Grouping by behaviour (flat, sharp drop, reversal) keeps a busy figure readable.
ANSWER THEY GOT **23 solutes reverse direction** Er: 5.6 → 10.2 mJ/m²	For Cr, Cu, Dy, Er, Fe, Gd, Y, Zr and others, the SFE moves back toward pure Mg at 25 at.%; this holds for all rare earths except La, Ce, Pr, Nd and Yb.	An unexpected result deserves its own paragraph and a tentative explanation, here quantum tunneling.
ANSWER THEY GOT **Bi and Yb drop sharply** Bi: 13.9 → 2.7 mJ/m²	These solutes lower the SFE much more at the higher concentration, which the authors suggest may raise stacking-fault density.	Turning a result into a testable suggestion for experiments adds value.

R5 · Links Results to Introduction step 8

What do the results deliver?

Table 2 is the promised SFE database, and the raise/lower split sets up the physical explanation in §4.1.

> ***Writing move.*** *End the results by tying findings to the promises made in the introduction. A reader should be able to tick off each promise.*

NODE	IN THE PAPER	WRITING MOVE
THIS WORK **The database is Table 2** 52 solutes × 2 concentrations	The full table of SFEs, with ionization energies and radii, is the promised reference for alloy design.	Point explicitly back to the promise when you deliver it.
THIS WORK **Trend sets up the discussion** §4.1: low ionization energy + large radius → low SFE	The raise/lower split lines up with ionization energy and atomic radius, which the discussion then explains with charge-density plots.	End results with the pattern the discussion will explain, so the next section feels necessary.
THIS WORK **Agrees with Mg–Y experiments** rare-earth trend vs [6]	The rare-earth result matches the experimental finding that Y lowers the I_1 SFE and improves ductility.	Connecting back to the cited evidence in the introduction closes the argument.

R6 · Not in the paper: a reader's check

What should a reader double-check?

A few points in the results are inconsistent, untested or missing a comparison a reviewer would ask for.

> ***Writing move.*** *Before submitting, re-read your results against your own tables and against the gap you claimed in the introduction.*

NODE	IN THE PAPER	WRITING MOVE
NOT REPORTED OR UNCLEAR **Ta appears in both groups** Table 2: 21.7 · 24.6 mJ/m² → raises	Tantalum is listed among the SFE-raising and the SFE-lowering elements. Table 2 shows it raises the SFE.	Check every list in the text against your own table.
NOT REPORTED OR UNCLEAR **No comparison with earlier I_1 values** Al, Dy, Gd, Li, Nd, Pr, Tb, Y [7,8,21]	The introduction names eight solutes already computed by others, but the results never compare against them.	If your gap mentions prior values, compare with them; it is the most direct test of your new setup.
NOT REPORTED OR UNCLEAR **Negative SFE not followed up** stability only suggested	The paper suggests the faulted structure may be stable for nine solutes but does not test it.	Flag such claims as open questions or add a quick check.
NOT REPORTED OR UNCLEAR **No uncertainty on small changes** e.g. Mo 21.6 vs 21.0 · Rh 21.3 vs 21.0	Differences under about 1 mJ/m² are counted as increases, though the pure-Mg values already vary by more than that across cells.	State a numerical uncertainty and call changes inside it "no significant effect".

Why it happens

The discussion names the results that need explaining, proposes two simple descriptors, tests them against the data, explains the exceptions, and ends with what it means for alloy design.



Grey lines link each node to its step. Green arrows: a gap answered, a promise delivered, or a puzzle explained. Dashed blue arrows: an idea planted in one place and used in another.

D0 · Links Results to Discussion

Which results need explaining?

Three findings call for an explanation: why some solutes raise and others lower the SFE, why every rare earth lowers it, and why 23 solutes reverse direction at higher concentration.

> ***Writing move.*** *Open the discussion by naming the results you will explain. Do not re-report them; point to them.*

NODE	IN THE PAPER	WRITING MOVE
ANSWER THEY GOT **Raise vs lower split** 18 raise · most lower	Solutes divide into SFE raisers and lowerers.	The main pattern in the results is the first thing to explain.
ANSWER THEY GOT **Every rare earth lowers SFE** no exceptions	An exceptionless pattern suggests a single underlying cause.	Strong regularities are the most rewarding results to explain.
ANSWER THEY GOT **23 solutes reverse at 25 at.%** e.g. Er: 5.6 → 10.2 mJ/m²	For these solutes more solute does not mean a larger change.	An anomaly needs its own explanation, which comes in D7.

D1 · Section 4.1, paragraphs 1–2

Which properties could control SFE?

Solutes disturb both the electrons and the strain field around them, so the authors propose two descriptors: first ionization energy and atomic radius.

> ***Writing move.*** *State the hypothesis and why it is plausible before testing it. Link it to the mechanism you planted in the introduction.*
>
> *Signal phrase: "In this section, we try to find the general trends"*

NODE	IN THE PAPER	WRITING MOVE
CONCEPT OR CHOICE **Electronic effect → 1st ionization energy** E_i of Mg = 738 kJ/mol	How easily a solute gives up an electron is used as a proxy for how it disturbs the local charge density.	Pick a descriptor that is tabulated for every element so the rule can be applied widely.
CONCEPT OR CHOICE **Strain effect → atomic radius** r of Mg = 160 pm	Solute size is used as a proxy for local lattice strain.	A second, independent descriptor covers what the first misses.
CONCEPT OR CHOICE **Tested on 11 at.% data** limits solute–solute interaction	The lower concentration is used so each solute acts roughly on its own.	Say which subset of data a test uses, and why.

D2 · Section 4.1, paragraph 3, Fig. 4

Does ionization energy predict it?

Almost every SFE-lowering solute has a lower first ionization energy than Mg, and most with a higher one raise the SFE. Ten solutes break the rule.

> ***Writing move.*** *Test one descriptor at a time against the data, and name the exceptions openly instead of hiding them.*
>
> *Signal phrase: "It is interesting to note that"*

NODE	IN THE PAPER	WRITING MOVE
CITED SOURCE **Fig. 4: SFE vs ionization energy** 52 solutes at 11 at.%	A scatter plot with Mg marked as the reference point.	One plot per descriptor keeps each test clear.
ANSWER THEY GOT **Mostly yes** E_i below Mg → lower SFE	Nearly all lowerers have lower ionization energy than Mg; solutes above it raise the SFE.	Report how well the rule works before the exceptions.
ANSWER THEY GOT **Ten exceptions** Ag Cr Li Mn Mo Nb Ni Rh Ru V	These have lower ionization energy than Mg yet raise the SFE.	Listing exceptions by name sets up the next test.

D3 · Section 4.1, paragraph 4, Fig. 5

Does atomic radius predict it?

Virtually all SFE-raising solutes are smaller than Mg, including the ten that broke the ionization-energy rule.

> ***Writing move.*** *Use the second descriptor to account for the failures of the first. Highlighting the exceptions in the figure (red labels) makes the point visible.*

NODE	IN THE PAPER	WRITING MOVE
CITED SOURCE **Fig. 5: SFE vs atomic radius** exceptions in red	The ten ionization-energy exceptions are labelled in red.	Visual highlighting connects two figures for the reader.
ANSWER THEY GOT **SFE raisers are smaller than Mg** r < 160 pm	Virtually every SFE-raising solute has a smaller radius than Mg.	State the second correlation as plainly as the first.
ANSWER THEY GOT **All ten exceptions are small atoms** 124–152 pm	Size accounts for the solutes that ionization energy could not.	This is the key move of the discussion: the second descriptor rescues the first.

D4 · Section 4.1, paragraphs 5–6, Fig. 6

What happens to the electrons?

Charge-density plots show Co drawing electrons toward itself and Cs pushing them away. More charge near the fault means more redistribution is needed to form it, so a higher SFE.

> ***Writing move.*** *Move from correlation to mechanism with a direct picture of the physics, using the two most extreme cases.*

NODE	IN THE PAPER	WRITING MOVE
CITED SOURCE **Charge-density plots** Mg, Mg–Co, Mg–Cs · VESTA · iso 0.015	Iso-surfaces from the VASP charge density show where electrons gather around each solute.	Choose the extreme cases for the clearest picture, and report how the plot was made.
CONCEPT OR CHOICE **Potential well or hill** [33] Myers	A solute with higher ionization energy creates a local potential well that gathers electrons; a lower one creates a hill that repels them.	A textbook concept bridges the descriptor and the plots.
CONCEPT OR CHOICE **Co gathers electrons → higher SFE** more charge to redistribute	Higher charge density near the doped plane makes forming the fault cost more energy.	Spell out the causal chain step by step.
CONCEPT OR CHOICE **Cs repels electrons → lower SFE** less charge near the fault	Lower charge density near the doped plane makes the fault cheaper to form.	The mirror case confirms the chain works in both directions.

D5 · Section 4.1, paragraph 7, with Fig. 1

Why does atom size matter?

The fcc-like zone at the fault is slightly wider than the perfect lattice. Large solutes expand the lattice locally, which favours forming that wider zone and lowers the SFE.

> ***Writing move.*** *Explain each descriptor with its own physical picture, here a structural one drawn from the methods.*

NODE	IN THE PAPER	WRITING MOVE
NUMBER OR SETTING **The fault zone is wider** 5.178 Å vs 5.165 Å	After relaxation, the interplanar distance in the fcc-like zone exceeds that of the perfect lattice.	A small number from the methods becomes the basis of an argument.
CONCEPT OR CHOICE **Large solutes favour the wider zone** local lattice expansion	A large solute already expands the lattice, so forming the wider fault zone is easier and the SFE drops.	Connect the structural fact to the size correlation.

D6 · Section 4.1, final paragraph

What is the combined rule?

Ionization energy and atomic radius are two competing factors. Solutes with low ionization energy and large radius lower the I_1 SFE.

> ***Writing move.*** *Condense the discussion into one rule a reader can carry away and apply to an element you did not compute.*
>
> *Signal phrase: "two competitive factors"*

NODE	IN THE PAPER	WRITING MOVE
THIS WORK **Low E_i + large radius → low SFE** two competing factors	The two descriptors together predict whether a solute lowers or raises the I_1 SFE.	The take-home rule of the paper, stated in one line.
ANSWER THEY GOT **Common alloying elements fit** Al, Ca, Sr, Zr, rare earths	Elements already used in commercial Mg alloys all have lower ionization energy than Mg and lower the SFE.	Showing the rule covers familiar cases makes it credible to practitioners.

D7 · Section 4.2, final paragraph

Why do some reverse at 25 at.%?

Above about 20 at.%, electrons may tunnel between neighbouring solute atoms, weakening the local potential wells and hills, so the SFE change shrinks. The authors call this preliminary.

> ***Writing move.*** *Offer a tentative explanation for an unexpected result, cite its source, and say clearly how certain you are.*
>
> *Signal phrase: "needs further verifying"*

NODE	IN THE PAPER	WRITING MOVE
CONCEPT OR CHOICE **Electron tunneling above ~20 at.%** [33] Myers	At high solute concentration, electrons can tunnel between solute atoms, and their interaction grows.	Name the source of a borrowed idea and the threshold it predicts.
CONCEPT OR CHOICE **Weaker wells and hills** 25 at.% is above the threshold	Fewer electrons are drawn in or pushed away, so the SFE change is smaller than expected.	Show how the idea produces the observed anomaly.
CITED SOURCE **Matches optimum rare-earth additions** [34–37]	Experiments often find the best effect at a low rare-earth content, consistent with non-monotonic SFE.	Experimental agreement supports a tentative explanation.
GAP **Authors: mechanism needs verifying** stated in §4.2	The paper calls its tunneling explanation preliminary.	Honest hedging protects credibility when evidence is thin.

D8 · Sections 3.2, 4.2 and 5

What does it mean for alloy design?

The rule and the rare-earth trend point to a mechanism for improving Mg ductility, and the concentration results suggest that an optimum solute amount exists.

> ***Writing move.*** *Return to the problem from the first paragraph of the introduction. The discussion should end where the paper began.*

NODE	IN THE PAPER	WRITING MOVE
THIS WORK **A screening rule for Mg alloys** choose low-E_i, large solutes	The rule lets designers pre-select solutes likely to lower the SFE and aid non-basal slip.	Translate the science back into the practical problem from the introduction.
THIS WORK **Experiments to try next** Yb, Bi · rare-earth optimum	Yb and Bi may raise stacking-fault density, and rare-earth content should be optimised rather than maximised.	Ending with testable suggestions invites follow-up work.

D9 · Not in the paper: a reader's check

What should a reader question?

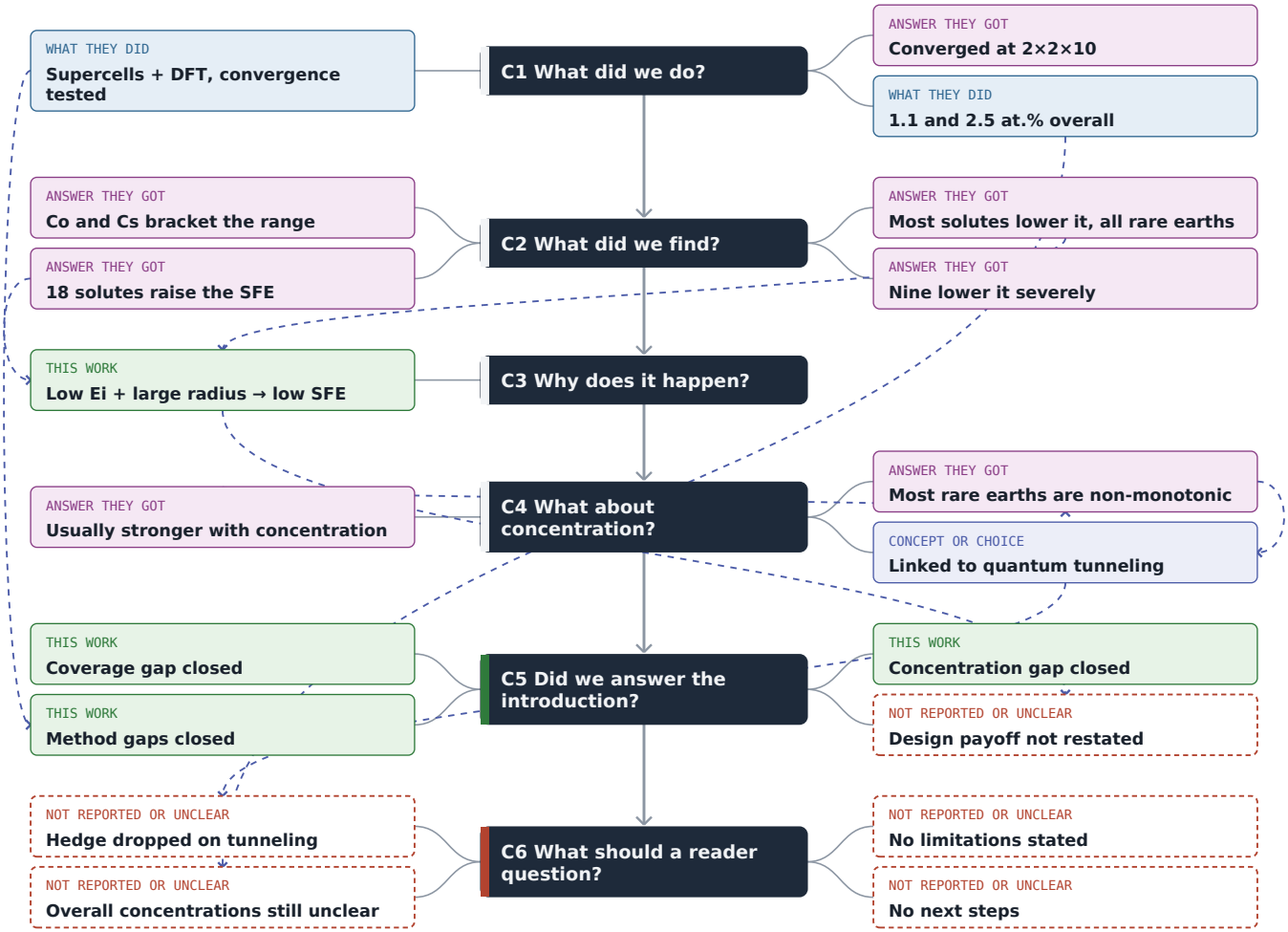
The descriptors are correlations, the mechanism rests on two pictures, and the concentration explanation is untested.

> ***Writing move.*** *For each explanation in your own discussion, ask what evidence would convince a sceptic, and whether you have shown it.*

NODE	IN THE PAPER	WRITING MOVE
NOT REPORTED OR UNCLEAR **Correlation, not proof** free-atom descriptors	Ionization energy and radius describe isolated atoms. The trends are correlations; no fit or count of violations is given.	Quantify how well a rule works, for example how many of 52 solutes it predicts correctly.
NOT REPORTED OR UNCLEAR **Mechanism shown for two solutes** Co and Cs only	The electron picture rests on two plots and no quantitative charge analysis.	Back pictures with a number (charge transfer for each solute) to make a mechanism convincing.
NOT REPORTED OR UNCLEAR **Tunneling not tested** alternative: solute images ~6.4 Å apart	In the 2×2 cell, each solute sits two lattice spacings from its periodic image, so direct solute–solute interaction or strain overlap could also explain the reversal.	Consider alternative explanations and say why yours is preferred.
NOT REPORTED OR UNCLEAR **No weighting of the two factors** "competitive" but not quantified	The paper does not say which factor dominates when they disagree.	If two factors compete, show how they combine, even with a simple map of E_i against radius.

What the paper claims at the end

Each sentence of the one-paragraph conclusion is a node. Step C5 checks the introduction's gaps and promises one by one; step C6 flags where the conclusion says more, or less, than the paper showed.



Grey lines link each node to its step. Green arrows: a gap answered, a promise delivered, or a puzzle explained. Dashed blue arrows: an idea planted in one place and used in another.

C1 · Section 5, sentences 1–3

What did we do?

A series of supercells was built for DFT, convergence was tested strictly, and alloys were computed at two concentrations in the 3×3×10 and 2×2×10 cells.

> ***Writing move.*** *Open the conclusion with one or two sentences on the approach, enough for a reader who skipped straight to the end.*
>
> *Signal phrase: "In this paper, a series of supercells were constructed"*

NODE	IN THE PAPER	WRITING MOVE
WHAT THEY DID **Supercells + DFT, convergence tested** §5 sentences 1–2	The conclusion opens with the method and stresses that convergence was tested strictly.	Leading with rigour reminds the reader why to trust the numbers.
ANSWER THEY GOT **Converged at 2×2×10** cf. ≥ 10 unit cells in §3.1	A satisfactory convergence is reported for the 2×2×10 cell.	Restate the validation result in one clause.
WHAT THEY DID **1.1 and 2.5 at.% overall** = 11 and 25 at.% on the fault plane	The two concentrations are given both as overall and as fault-plane values.	Giving both numbers helps readers compare with experiments, but they must match the methods.

C2 · Section 5, sentences 4–5

What did we find?

Co and Cs bracket the range of SFE. Most solutes, including every rare earth, lower the SFE; nine lower it severely; eighteen raise it.

> ***Writing move.*** *Restate the key numbers and patterns, not the whole table. The extremes and the strongest pattern are enough.*

NODE	IN THE PAPER	WRITING MOVE
ANSWER THEY GOT **Co and Cs bracket the range** 54.1 to −33.3 mJ/m²	Cobalt raises the SFE most and cesium lowers it most.	Two numbers give the full scale of the effect.
ANSWER THEY GOT **18 solutes raise the SFE** Ag Be Co Cr Cu Fe Li Mn Mo Nb Ni Pd Re Rh Ru Ta V W	The raising group is listed in full, with Ta correctly included.	Lists in the conclusion must match the results; here the conclusion is right where §3.2 was not.
ANSWER THEY GOT **Most solutes lower it, all rare earths** majority of 52	The strongest pattern, every rare earth lowering the SFE, is restated.	Repeat the finding readers are most likely to remember and cite.
ANSWER THEY GOT **Nine lower it severely** Ba Ce Cs K La Nd Pr Rb Yb	These are the solutes whose SFE turned negative.	"Severely" softens "negative"; the conclusion could say what negative SFE implies.

C3 · Section 5, sentence 6

Why does it happen?

Solutes with lower ionization energy and larger radius decrease the SFE; those with higher ionization energy and smaller radius increase it.

Writing move. *Give the take-home rule in one sentence, in the same words as the discussion.*

NODE	IN THE PAPER	WRITING MOVE
THIS WORK **Low E_i + large radius → low SFE** and the reverse	The combined rule from the discussion, stated in both directions.	The clearest sentence of the conclusion: a rule others can reuse.

C4 · Section 5, sentences 7–10

What about concentration?

The effect usually grows with concentration, but some solutes, including most rare earths, vary non-monotonically, which the paper links to quantum tunneling.

Writing move. *Report secondary findings after the main one, and keep the same level of certainty as in the discussion.*

NODE	IN THE PAPER	WRITING MOVE
ANSWER THEY GOT **Usually stronger with concentration** e.g. Ga	For most solutes the SFE change grows from 11 to 25 at.%.	State the general trend before the exceptions.
ANSWER THEY GOT **Most rare earths are non-monotonic** all except La, Ce, Pr, Nd, Yb	The rare-earth exceptions to the general trend are singled out.	Linking this to optimum rare-earth additions in experiments would strengthen it, as §4.2 did.
CONCEPT OR CHOICE **Linked to quantum tunneling** §5 final sentence	The final sentence says the abnormal variation was correlated with quantum tunneling.	Ending on the least certain claim weakens the close; ending on the design payoff would be stronger.

C5 · Compares Section 5 with Introduction steps 6–8

Did we answer the introduction?

Three of the four gaps from the introduction are clearly closed. The practical payoff promised in the introduction is not restated at the end.

> ***Writing move.*** *Read your conclusion next to your gap and "this work" paragraph. Every promise should be visibly kept, and the opening problem should return.*

NODE	IN THE PAPER	WRITING MOVE
THIS WORK **Coverage gap closed** 8 solutes before → 52 now	The introduction said only eight solutes had I_1 values; the paper delivers 52.	A clear, quantifiable answer to the gap.
THIS WORK **Method gaps closed** no vacuum · convergence tested	The two-fault supercell and the convergence test answer both method gaps.	The conclusion mentions convergence but not the no-vacuum design, the paper's main methodological novelty.
THIS WORK **Concentration gap closed** two concentrations	The implicit single-concentration gap is answered with 11 and 25 at.%.	Stating this gap in the introduction would make this answer count for more.
NOT REPORTED OR UNCLEAR **Design payoff not restated** database · alloy design · ductility	The introduction promised a database to guide alloy design and opened with Mg formability. Neither returns in the conclusion.	End where you began: one sentence linking the findings back to the original problem.

C6 · Not in the paper: a reader's check

What should a reader question?

The conclusion drops a hedge from the discussion, states no limitations and suggests no next steps.

> ***Writing move.*** *A conclusion should claim exactly what the paper showed: no more certain than the discussion, and with its limits stated.*

NODE	IN THE PAPER	WRITING MOVE
NOT REPORTED OR UNCLEAR **Hedge dropped on tunneling** §4.2 "needs further verifying" → §5 stated plainly	The discussion calls the tunneling explanation preliminary; the conclusion presents it without that caveat.	Carry hedges through to the conclusion; claims should not grow stronger at the end.
NOT REPORTED OR UNCLEAR **Overall concentrations still unclear** 1/80 = 1.25 % · 1/180 = 0.56 %	The 1.1 and 2.5 at.% figures repeat the ambiguity in the methods about one solute per cell versus one per fault.	Numbers in the conclusion should be derivable from the methods.
NOT REPORTED OR UNCLEAR **No limitations stated** one fault type · one solute site · 0 K · spin	Only the I_1 fault, one solute position, ground-state DFT and an unstated spin treatment are used, but none of these limits are mentioned.	Two sentences on limitations make a conclusion more credible, not less.
NOT REPORTED OR UNCLEAR **No next steps** Yb, Bi experiments · tunneling test	The discussion suggested experiments and admitted an untested mechanism, but the conclusion proposes no future work.	Close with what should be done next, so readers know where to build on your work.

Across the paper

The golden thread

Each row follows one item from the introduction through the rest of the paper. A strong paper lets you fill every cell. Empty or weak cells show where the argument loses its grip.

THREAD	INTRODUCTION GAP	PROMISE	METHOD	RESULT	CONCLUSION
Coverage gap	Only 8 solutes had I_1 values	52 binary Mg–X systems	One Mg replaced by a solute in the faulted layer	18 raise, most lower; Table 2	Restated in full
Vacuum-layer gap	Free surfaces may interfere with the fault	Two-fault periodic supercell	Two interfaces per cell, no vacuum	Pure Mg agrees with published DFT	Not mentioned
Convergence gap	Size dependence rarely tested	Convergence test	15 supercells, 1×1 to 3×3 × 4–12	Stable at ≥ 10 unit cells	“Strictly tested”
Concentration gap (implicit)	Never stated	Two concentrations	1 of 4 and 1 of 9 atoms on the plane	Usually stronger at 25 at.%; 23 reverse	Restated; tunneling hedge dropped
Design payoff	Mg formability (intro step 2)	SFE database for alloy design	n/a	Table 2	Not restated
Physical trends	Alloying tunes SFE (intro step 3)	Trends from simple features	n/a	Raise/lower split	Rule restated in both directions

Shading in the last column: green, clearly closed; amber, closed but weakly reported; red, promised but not delivered at the end.

What the thread shows

- The paper is strongest where its gaps are explicit and quantifiable. “Eight solutes before, 52 now” is visible from the first page to the last.
- Its main methodological novelty, the no-vacuum supercell, disappears from the conclusion. Readers who skim the abstract and conclusion may miss what makes the method different.
- The concentration study answers a gap the introduction never states. Naming that gap would make the result count for more.
- The opening problem, poor formability of Mg, never returns. One closing sentence linking the rule back to ductility would complete the circle.

For your own paper

Reader's checklist

These are the points a careful reader or reviewer would raise about this paper. Each one is phrased as a check you can run on your own manuscript before submission.

SECTION	ISSUE IN THIS PAPER	CHECK FOR YOUR PAPER
Introduction	**Only one solute concentration** Not stated outright, but the contribution of two concentrations only makes sense if earlier work used one.	☐ An unstated gap weakens the argument. Writing it out would make step 7 land harder.
Methods	**Plane-wave cutoff energy** The basis-set cutoff is not reported, although it directly affects total energies.	☐ Always report the cutoff and how it was tested.
	Spin polarization The text does not say whether calculations were spin-polarized. For magnetic solutes this can change the energy noticeably.	☐ State the spin treatment whenever transition-metal solutes are involved.
	Lattice parameters and relaxation scope The Mg lattice constants used, and whether the cell shape or only atom positions were relaxed, are not given.	☐ Report the equilibrium lattice constants and the degrees of freedom that were relaxed.
	Overall concentrations unclear The paper gives 2.5 and 1.1 at.% overall. Those match two solutes per cell, while the text describes replacing one atom. It may mean one per fault interface, but this is not stated.	☐ Check that every number in the methods can be derived from the description.
Results	**Ta appears in both groups** Tantalum is listed among the SFE-raising and the SFE-lowering elements. Table 2 shows it raises the SFE.	☐ Check every list in the text against your own table.
	No comparison with earlier I_1 values The introduction names eight solutes already computed by others, but the results never compare against them.	☐ If your gap mentions prior values, compare with them; it is the most direct test of your new setup.
	Negative SFE not followed up The paper suggests the faulted structure may be stable for nine solutes but does not test it.	☐ Flag such claims as open questions or add a quick check.
	No uncertainty on small changes Differences under about 1 mJ/m² are counted as increases, though the pure-Mg values already vary by more than that across cells.	☐ State a numerical uncertainty and call changes inside it "no significant effect".
Discussion	**Correlation, not proof** Ionization energy and radius describe isolated atoms. The trends are correlations; no fit or count of violations is given.	☐ Quantify how well a rule works, for example how many of 52 solutes it predicts correctly.
	Mechanism shown for two solutes The electron picture rests on two plots and no quantitative charge analysis.	☐ Back pictures with a number (charge transfer for each solute) to make a mechanism convincing.
	Tunneling not tested In the 2×2 cell, each solute sits two lattice spacings from its periodic image, so direct solute–solute interaction or strain overlap could also explain the reversal.	☐ Consider alternative explanations and say why yours is preferred.

SECTION	ISSUE IN THIS PAPER	CHECK FOR YOUR PAPER
	No weighting of the two factors The paper does not say which factor dominates when they disagree.	☐ If two factors compete, show how they combine, even with a simple map of E_i against radius.
Conclusions	**Design payoff not restated** The introduction promised a database to guide alloy design and opened with Mg formability. Neither returns in the conclusion.	☐ End where you began: one sentence linking the findings back to the original problem.
	Hedge dropped on tunneling The discussion calls the tunneling explanation preliminary; the conclusion presents it without that caveat.	☐ Carry hedges through to the conclusion; claims should not grow stronger at the end.
	No limitations stated Only the I_1 fault, one solute position, ground-state DFT and an unstated spin treatment are used, but none of these limits are mentioned.	☐ Two sentences on limitations make a conclusion more credible, not less.
	No next steps The discussion suggested experiments and admitted an untested mechanism, but the conclusion proposes no future work.	☐ Close with what should be done next, so readers know where to build on your work.
	Overall concentrations still unclear The 1.1 and 2.5 at.% figures repeat the ambiguity in the methods about one solute per cell versus one per fault.	☐ Numbers in the conclusion should be derivable from the methods.

Template

Questions to answer in each section

The spine of each map, stripped of this paper’s content. Write one or two sentences answering each question before you draft; if a question has no answer, that is the part of the argument to fix first.

Introduction

1. Why does the field matter?
2. What is the specific problem?
3. What controls the problem?
4. Why believe it matters in this system?
5. How can it be studied, and why this method?
6. What is still missing? (coverage and method gaps)
7. What do we do? (one item per gap)
8. Why does it matter? (payoff and main finding)

Methods

1. Which theory, code and settings?
2. How is the system or sample built?
3. How is the quantity computed or measured?
4. What equation defines the result?
5. Is it converged and validated on a known case?
6. How is the variable of interest controlled?
7. Could a reader reproduce it from the text alone?

Results

1. Does the method reproduce known values?
2. What is the main split or trend in the data?
3. What are the extremes, with numbers?
4. How does the result depend on the controlled variable?
5. Which introduction promises do the results deliver?

Discussion

1. Which results need explaining?
2. What hypothesis could explain them, and why?
3. Does the hypothesis fit the data? Which cases break it?
4. What mechanism links cause and effect?
5. What single rule can a reader take away?
6. What alternative explanations exist, and why is yours preferred?

Conclusions

1. What did we do, in one sentence?
2. What did we find, with key numbers?
3. Why does it happen, in one rule?
4. Is every gap and promise from the introduction answered?
5. What are the limits, and what comes next?
6. Does the last sentence return to the opening problem?

Appendix A

How this analysis was made

The maps, checks and links in this report were produced in four stages. Each section was handled the same way, and the interactive page and this PDF are generated from one shared dataset.

Stage 1 · Fix a question spine for each section

Before extracting anything, each section was given an ordered set of questions based on how that kind of section usually argues, not on this paper.

SECTION	SPINE	BASIS
Introduction	8 steps: field → problem → key concept → evidence → methods landscape → gap → this work → payoff	Swales' CARS model (territory, niche, occupying the niche)
Methods	7 steps: theory and code → structure model → numerical settings → defining equation → validation → alloy model → reproducibility	Usual order of a computational methods section
Results	6 steps: one question of the data per step, each answered by a "what they did" and an "answer" node	Question-and-answer reading of results
Discussion	10 steps: puzzles → hypothesis → test 1 → test 2 → mechanism → combined rule → anomaly → payoff → checks	Explanatory argument structure
Conclusions	6 steps: recap of did / found / why / concentration, then a scorecard and checks	Recap mirrored against the introduction

Stage 2 · Extract statements into the spine

For each step, the relevant sentences were turned into typed nodes. About 125 statements were extracted. Each node records:

- a type: concept, cited source, number or setting, gap, this work, what they did, answer, or not reported;
- a paraphrase of what the paper says, and the writing move the statement performs;
- its references and its location in the paper (section, paragraph, figure or table).

Stage 3 · Check each part locally

Every section after the introduction ends with a reader's-check step, filled by three kinds of test:

TEST	QUESTION ASKED	EXAMPLE FOUND IN THIS PAPER
Completeness	Is everything needed to reproduce or judge this part reported?	Cutoff energy, spin treatment and lattice parameters are not given.
Arithmetic	Can every stated number be derived from the description?	One solute in 80 or 180 atoms gives 1.25 and 0.56 at.%, not the stated 2.5 and 1.1 at.%.
Consistency	Does the text agree with the paper's own tables and earlier sections?	Ta is listed among both SFE-raising and SFE-lowering elements.

Stage 4 · Link the parts

Two kinds of edges connect statements across sections. **Same-item** links join two nodes describing the same thing, for example a promise in the introduction and its twin at the top of the methods map. **Delivery** links show a gap answered by a promise, a promise delivered by a method, or a puzzle explained by a mechanism. The golden thread table and conclusion step C5 summarise these links.

Rendering and quality checks

All nodes and edges are stored in one data block. The interactive web page draws the maps from it; for this report the same data was exported to JSON and typeset with ReportLab, with map figures drawn from the data. Before publishing, the script syntax was checked, every node identifier was confirmed unique, every edge was confirmed to point to an existing node, and the rendered pages were inspected.

Appendix B

Comparison with the divide-and-conquer approach

The target approach analyses each part of a paper separately before assessing the whole-paper logic: the introduction first (background, gap and hypothesis), then each further part with local checks, then the links between parts. Statements carry a `paper_part` field so extraction can run part by part, and `render.py` lists statements per part. The table below compares this analysis with that approach.

REQUIREMENT	STATUS	COMMENT
Introduction first	**Partly**	The introduction was mapped first, but not in isolation: the whole paper had been read beforehand, so later sections shaped it. The implicit “one concentration” gap, for instance, was inferred from the contributions.
Extract background, gap and hypothesis	**Partly**	Background and gap are explicit. No hypothesis node exists, because the paper states aims rather than a hypothesis; the aims are captured as “this work” nodes.
Each further part, checked locally	**Mostly**	Methods, results, discussion and conclusion each have local checks. The paper has no theory section or SI. The abstract was not analysed.
Whole-paper logic	**Partly**	Traceability exists (golden thread, scorecard C5) and one certainty-drift check (tunneling). There is no explicit verdict on whether each conclusion follows from the hypothesis and results, and no score.
paper_part on every statement	**Different**	Nodes are grouped by function, not by location. The §3.1 validation sits in the methods map and the §4.2 concentration results in the results map, whereas paper_part records where a statement appears.
render.py lists statements per part	**No**	The target format and render.py were not available, so a separate renderer was written. The data could be exported to the target schema.

Two changes would close most of the gap

- **Record both location and function.** Keep `paper_part` for where a statement appears and add a field such as `role` for what it does. A mismatch between the two is itself a finding: results reported in the discussion are exactly the misplaced-result issue noted for §4.2.
- **Make the introduction pass blind.** Give the first pass only the introduction (and optionally the abstract), and have it record the hypothesis, aims and promised payoff before any other part is read. Later parts are then judged against a fixed promise rather than one shaped by hindsight.

What was tried

The approach was applied by hand to Zhang et al. (2013). Each section was mapped onto a fixed question spine (introduction 8 steps, methods 7, results 6, discussion 10, conclusion 6), with about 125 typed statements. Local checks found 17 issues: 4 unreported method settings, 4 results inconsistencies or missing comparisons, 4 discussion weaknesses and 5 conclusion problems. Linking the parts showed that three of four introduction gaps are closed, the main methodological novelty is not mentioned in the conclusion, the design payoff is never restated, and one claim

(tunneling) becomes more certain between discussion and conclusion. Limitations: the introduction was analysed after reading the whole paper, so it was not independent, and no hypothesis was extracted because the paper states aims rather than a hypothesis.

Open questions and suggested answers

Should each part get its own sub-score before the whole-paper score? Yes, but keep sub-scores separate rather than averaging them into the whole-paper score. Local quality (completeness, internal consistency, reproducibility) and argument quality (whether the parts connect) fail in different ways. This paper shows why: its methods are clear and its results internally sound, yet the whole-paper logic has gaps that a part average would hide. A simple scheme is a count of issues by severity per part, with the whole-paper score driven by the link checks.

How should the introduction's hypothesis be compared with what the conclusion claims? Treat it as matching two lists. From the introduction, extract promise records (gap, aim or hypothesis, expected payoff); from the conclusion, extract claim records. Then classify each pair:

OUTCOME	DEFINITION	EXAMPLE IN THIS PAPER
Delivered	A claim matches the promise at the same strength.	Coverage gap: 8 solutes before, 52 now.
Dropped	A promise has no matching claim.	The design payoff, and the no-vacuum method, are absent from the conclusion.
Strengthened	The claim is more certain than the evidence or the discussion.	Tunneling: "needs further verifying" in §4.2, stated plainly in §5.
Weakened	The claim is less than what was promised.	None clearly in this paper.
Unpromised	A claim with no matching promise.	Concentration effects answer a gap the introduction never states.

Storing a certainty level on every statement (for example: stated, hedged, speculative) makes the strengthened case detectable automatically, which is usually the kind of drift reviewers care about most.

Source: J. Zhang, Y. Dou, G. Liu, Z. Guo, Comput. Mater. Sci. 79 (2013) 564–569, doi:10.1016/j.commatsci.2013.07.012. Node text paraphrases the paper; reference numbers in brackets follow its reference list. Checks marked as not reported reflect what is absent from the published text, not necessarily from the authors' calculations.